INTELLIGENCE

Ciccarelli & White (2015). *Psychology in Action*. Pearson. [Chapter 7.4]

- Ability to learn from one’s experiences, acquire knowledge, use resources effectively in adapting to new situations or solving problems
- Includes reasoning, memory, creativity, processing speed

Not just “being smart” — it's **how effectively you use your mind**

(Sternberg & Kaufman, 1998; Wechsler, 1975)

KEY QUESTIONS

- Is intelligence one ability or many?
- Can it be measured?
- Nature vs nurture?
- Does IQ predict success?

SPEARMAN’S TWO FACTOR THEORY OF INTELLIGENCE

Proposed (~1904)

- g factor (general intelligence) – *ability to reason and solve problems*
- Specific intelligence (s) – *task-specific abilities in certain areas such as music, business, art etc.*

THURSTONE'S THEORY OF INTELLIGENCE (~1938)

- Intelligence = **multiple independent abilities**
  - Verbal comprehension
  - Word fluency
  - Numerical ability
  - Spatial ability
  - Memory
  - Inductive Reasoning
  - Perceptual Speed
- No single "g", more **modular thinking**

GUILFORD’S STRUCTURE OF INTELLECT (SOI) THEORY (~1967)

Intelligence = many independent abilities (not one IQ)

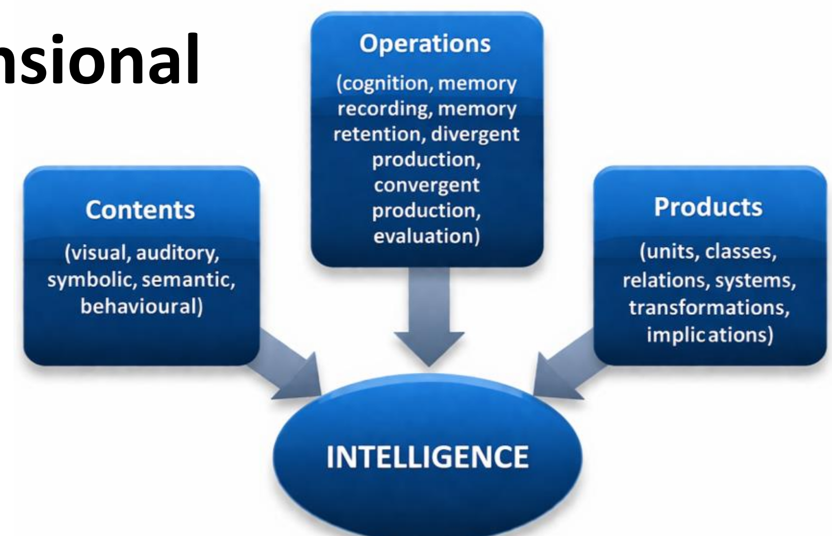
3 Dimensions - Operations, Contents, Products

Core Idea - Intelligence = **Operations × Contents × Products**

→ 6x5x6 = 180 specific abilities

Creativity **(divergent thinking)**

Shows intelligence is **multi-dimensional**



GARDNER'S THEORY OF MULTIPLE INTELLIGENCES

- Believed - *There are multiple intelligences with autonomous intelligence capacities.*

Gardner's Nine Intelligences

TYPE OF INTELLIGENCE	DESCRIPTION	SAMPLE OCCUPATION
Verbal/linguistic	Ability to use language	Writers, speakers
Musical	Ability to compose and/or perform music	Musicians, even those who do not read musical notes but can perform and compose
Logical/mathematical	Ability to think logically and to solve mathematical problems	Scientists, engineers
Visual/spatial	Ability to understand how objects are oriented in space	Pilots, astronauts, artists, navigators
Movement	Ability to control one's body motions	Dancers, athletes
Interpersonal	Sensitivity to others and understanding motivation of others	Psychologists, managers
Intrapersonal	Understanding of one's emotions and how they guide actions	Various people-oriented careers
Naturalist	Ability to recognize the patterns found in nature	Farmers, landscapers, biologists, botanists
Existentialist (a candidate intelligence)	Ability to see the "big picture" of the human world by asking questions about life, death, and the ultimate reality of human existence	Various careers, philosophical thinkers

Source: Gardner, 1998, 1999b.

STERNBERG'S TRIARCHIC THEORY OF INTELLIGENCE (~1996)

- 3 types of intelligence:
  - **Analytical** (problem-solving)
  - **Creative** (innovation)
  - **Practical** (street smart)
- Real-world success ≠ just IQ

EMOTIONAL INTELLIGENCE

- Ability to:
  - Understand emotions
  - Manage emotions
  - Empathize with others
- Important for:
  - Leadership
  - Relationships
  - Mental health

Salovey & Mayer, 1990; Dan Goleman, 1995

ASSESSMENT OF INTELLIGENCE

- What is IQ?

  IQ = Intelligence Quotient

  Originally: IQ = (Mental Age / Chronological Age) × 100

  Now: Based on **standardized scores**

TYPES OF TESTS

Binet’s mental ability test:

- Alfred Binet was asked by the French Ministry of Education to design a formal test of intelligence that would help identify children who were unable to learn as quickly
- Binet decided that the key element to be tested was a child’s **mental age**, or the average age at which children could successfully answer a particular level of questions

William Stern

- German psychologist William Stern gave a formula called IQ
- IQ = MA/CA * 100
- For example, if a child who is 10 years old and has a mental age of 15 (is able to answer the level of questions typical of a 15-year-old), his IQ would be 15/10 * 100 = 150

David Wechsler

Wechsler developed the Wechsler Adult Intelligence Scale (WAIS) and later the Wechsler Intelligence Scale for Children (WISC), an intelligence test for school-aged children.

- WAIS (Wechsler Adult Intelligence Scale: 16 years to 90 years and 11 months

- WISC (Wechsler Intelligence Scale for Children) : 7-16 years 11 subtests = separate scores for each area (Verbal IQ and Performance IQ).

- WISC has 11 subtests = separate scores for each area (Verbal IQ and Performance IQ). It is an Excellent tool for identifying learning disorders.

- WPPSI (Wechsler Preschool and Primary Scale of Intelligence (2 years 6months to 7 years 7 months)

WISC

VERBAL

General Information
What day of the year is Independence Day?

Similarities
In what way are *wool* and *cotton* alike?

Arithmetic Reasoning
If eggs cost 60 cents a dozen, what does 1 egg cost?

Vocabulary
Tell me the meaning of corrupt.

Comprehension
Why do people buy fire insurance?

Digit Span
Listen carefully, and when I am through, say the numbers right after me.

7 3 4 1 8 6

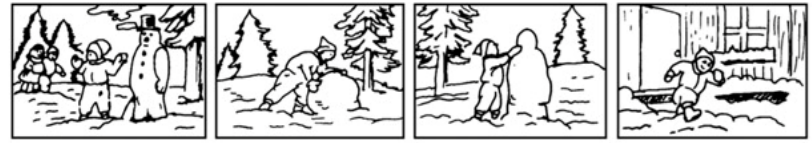
Now I am going to say some more numbers, but I want you to say them backward.

3 8 4 1 6

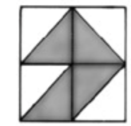
PERFORMANCE

Picture Completion
I am going to show you a picture with an important part missing. Tell me what is missing.

Picture Arrangement
The pictures below tell a story. Put them in the right order to tell the story.



Block Design
Using the four blocks, make one just like this.



Object Assembly
If these pieces are put together correctly, they will make something. Go ahead and put them together as quickly as you can.

Digit-Symbol Substitution

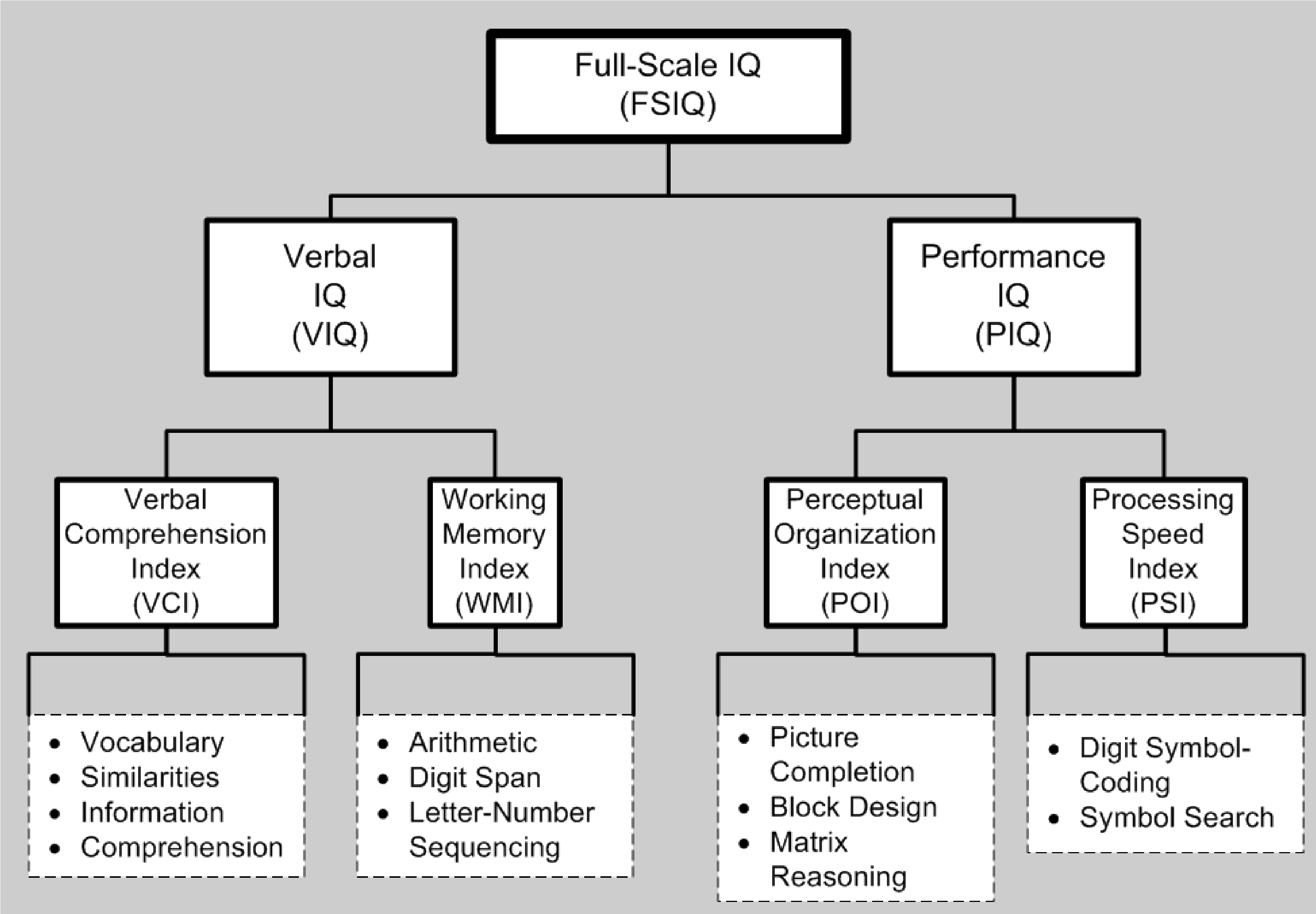
Code

△	○	▱	×	◇
1	2	3	4	5

Test

1	5	4	2	1	3	5	4	1	5

WAIS



Full-Scale IQ (FSIQ)
Verbal IQ (VIQ)
Performance IQ (PIQ)
Verbal Comprehension Index (VCI)
Working Memory Index (WMI)
Perceptual Organization Index (POI)
Processing Speed Index (PSI)
• Vocabulary
• Similarities
• Information
• Comprehension
• Arithmetic
• Digit Span
• Letter-Number Sequencing
• Picture Completion
• Block Design
• Matrix Reasoning
• Digit Symbol-Coding
• Symbol Search

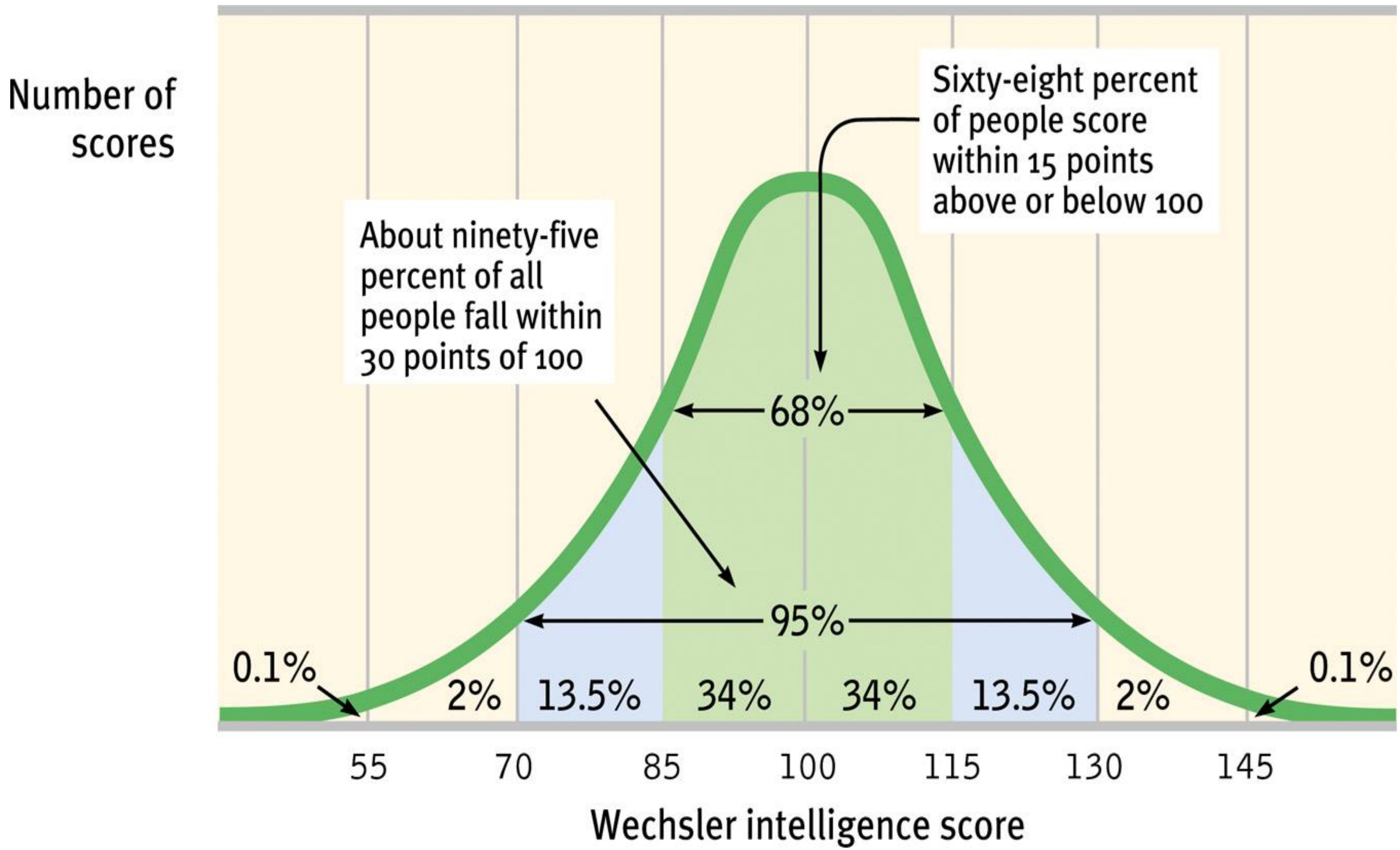
Table 2. WAIS-IV subtests, description, and CHC classification.

Subtests	Abbreviation	What it measures	CHC abilities
Verbal Comprehension Index			
Similarities	SI	Verbal concept formation, verbal reasoning, lexical knowledge, induction	Gc, Gf
Vocabulary	VC	Lexical knowledge, verbal concept formation	Gc
Information	IN	General knowledge	Gc
Perceptual Reasoning Index			
Block design	BD	Nonverbal concept formation and reasoning, organization, visual-motor coordination	Gv
Matrix reasoning	MR	General sequential reasoning, spatial ability, perceptual organization	Gf
Visual puzzles	VP	Nonverbal reasoning, analyze and synthesize abstract visual stimuli, spatial visualization	Gv
Working Memory Index			
Digit span	DS	WM, memory span, cognitive flexibility, mental alertness, attention, mental manipulation, visuospatial imaging	Gsm
Arithmetic	AR	WM, numeral reasoning, mental manipulation, concentration, attention, mental alertness	Gf, Gsm, Gq
Letter–number sequencing	LN	WM, sequential processing, mental manipulation, attention, concentration, memory span, cognitive flexibility	Gsm
Processing Speed Index			
Symbol search	SS	Visual motor coordination, cognitive flexibility, attention, concentration, psychomotor speed, short-term visual memory	Gs
Coding	CD	Visual motor coordination, visual scanning, cognitive flexibility, attention, concentration, psychomotor speed, short-term visual memory, learning	Gs
Cancellation	CA	Visual selective attention, vigilance, perceptual speed, visual-motor ability	Gs

Source. Lichtenberger & Kaufman, 2009; Wechsler, Coalson, & Raiford, 2008.

Note. WAIS-IV = Wechsler Adult Intelligence Scale–IV; CHC = Cattell–Horn–Carroll theory; Gc = Crystallized Intelligence; Gf = Fluid Reasoning; Gv = Visual Processing; WM = Working Memory; Gsm = Short-Term Memory; Gq = Quantitative Knowledge; Gs = Processing Speed.

Wechsler Intelligence Score



IQ Distribution

- Bell curve
- Below 70: low
- Above 130: gifted

DEGREES OF MENTAL RETARDATION

Level	Approximate Intelligence Scores	Percentage of Persons with Retardation	Adaptation to Demands of Life
Mild	50–70	85%	May learn academic skills up to sixth-grade level. Adults may, with assistance, achieve self-supporting social and vocational skills.
Moderate	35–50	10%	May progress to second-grade level academically. Adults may contribute to their own support by laboring in sheltered workshops.
Severe	20–35	3–4%	May learn to talk and to perform simple work tasks under close supervision but are generally unable to profit from vocational training.
Profound	Below 20	1–2%	Require constant aid and supervision.

Source: Reprinted with permission from the *Diagnostic and Statistical Manual of Mental Disorders,* Fourth Edition, text revision.

USES OF INTELLIGENCE TESTS

- **Educational**
  - Identify: learning disabilities, gifted students
  - Guide: curriculum design, interventions
- **Clinical**
  - Diagnose: intellectual disability, cognitive decline
  - Neuropsychological assessment
- **Occupational**
  - Job selection, aptitude testing
  - Performance prediction
- **Research**
  - Study cognitive processes
  - Brain–behavior relationships
  - Understand individual differences

NATURE VS NURTURE

- Genetics ≈ ~50% influence
- Environment matters:
  - Education
  - Nutrition
  - Socioeconomic status, etc.
- Intelligence = **interaction of both**

Discussion

- Is IQ enough?
- Can intelligence be improved?